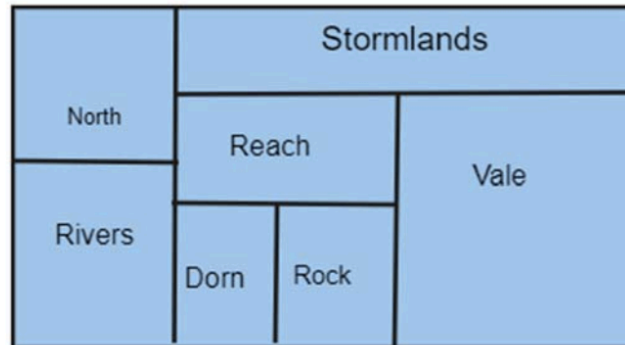


Practice problem (CSP)

Consider the following map of a Country where borders among the seven states have been marked. You need to color each of these divisions using three colors (Red/Green/Blue) such that two adjacent states do not receive the same color.

Now formulate the problem as CSP, Show the Constraint Graph, and solve the problem must applying both Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics. **[2.5+2.5]**



6. Consider the following map of Bangladesh where borders among the eight divisions have been marked. You need to color each of these divisions using three colors (Red/Green/Blue) such that two adjacent divisions do not receive the same color.

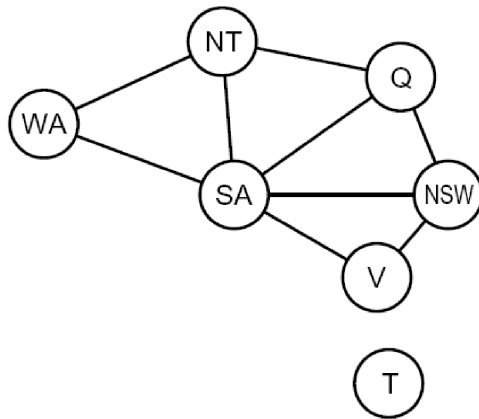
Now **formulate** the problem as CSP, Show the **Constraint Graph** and solve the problem **must applying** both Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics. **[5]**



For the given graph you have color all the nodes with red/green/blue so that the adjacent nodes don't contain same color:

i) Define the variables and domain for this problem

ii) Solve the problem as a CSP following **Minimum Remaining Value (MRV)** and **Least Constraining Value (LCV)** heuristics.



Sample Solution

i) Variables: WA, NT, Q, NSW, V, SA, T
Domain: {red, green, blue}

Step 1: Using MRV- all nodes have same number of remaining value (red, green value)
Break tie using degree heuristic, and choose SA.
Using LCV- all colors have the same impact, so choosing anyone (red).

SA←Red

Step 2: Using MRV- (WA, NT, Q, NSW, V) all have same number of remaining value (green, blue)
Break tie using degree heuristics, and then NT, Q, NSW have the same number of edges.
So, we can choose any, now choosing NT.
Using LCV- all colors have same impact, so choosing anyone (green)

NT←green

Step 3: Using MRV- (WA,Q) both have one color remaining.
Break the tie with using degree heuristic, and choose Q.
Q←blue (only blue is the option)

Step 4: Using MRV- (WA,NSW) both have one color remaining.
Break the tie with using degree heuristic, and choose NSW.
NSW←green (only green is the option)

Step 5: Using MRV- (WA,V) both have one color remaining.
The both have same degree (same number of edge), so choose any.
WA←B
V←B

Step 6: T is independent (no edge) so put any color
T←red